Malav Champaneria

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Education

Rutgers University — **Aug 2024 - May 2026**
Master of Science in Computer and Information Science (GPA = 3.85/4.00) — *New Jersey, USA*

Charotar University Of Science and Technology — **Aug 2020 - May 2024**
Bachelor of Technology in Information Technology (GPA = 9.02/10.00) — *Gujarat, India*

Technical Skills

Programming & Software Engineering: Python, SQL, JavaScript/TypeScript, C/C++, Bash, DSA, SDLC, OOP, FastAPI, Flask, REST APIs, Streamlit, Pydantic, Debugging, Testing, Version Control, uv, Poetry, ONNX, Deployment, Monitoring

AI/ML & GenAI: n8n, Agentic AI, Supervised & Unsupervised Learning, Deep Learning, GNNs, TensorFlow, Ensemble Methods, Hugging Face, NLP, Feature Engineering, Model Evaluation, Scikit-learn, Bert, PyTorch, RAG, FAISS, Embeddings, Prompt Engineering, Query Routing, LLMOps

Data & Analytics: EDA, Statistical Analysis, Data Cleaning, Feature Extraction, Structured & Unstructured Data Processing, ETL Pipelines, Data Modeling, JSON/CSV Processing

Systems, Robotics & Databases: Autonomous Navigation, RTK-GPS, RGB-Depth Camera, Sensor Fusion, PostgreSQL, MySQL, MS SQL, Oracle, Vector Databases, Query Optimization, System Integration

Visualization, Cloud & Tools: Matplotlib, Seaborn, Plotly, Tableau, PowerBI, AWS, Databricks, Docker, Git, CI/CD, Linux

Technical Experience

Research Assistant | *U.S. Department of Agriculture, New Jersey, USA* — **May 2025 - Aug 2025**

- Architected and deployed an end-to-end autonomous navigation pipeline for a Farm-ng robot in real farm environments, integrating GPS RTK (1–2 cm accuracy), RGB-depth vision, and ROS2-based control systems.
- Developed a 3D point cloud–based waypoint planning approach with depth-aware gradient heatmaps, improving navigation precision by ∼10% over RGB-only methods.
- Enabled reliable traversal across 10+ acres with 90% autonomous operation, reducing manual monitoring by 75% through robust sensor fusion and perception–control integration.

Python Developer | *INFLIBNET (Gov. Of India), Gandhinagar, Gujarat* — **Jan 2024 - Aug 2024**

- Integrated BERT and Sentence Transformers into an educational portal, boosting semantic relevance scores by 18%.
- Built metadata-driven keyword mapping, elevating precise query matches by 22% and lowering irrelevant results.
- Benchmarked retrieval models, with Sentence Transformers delivering 40% faster execution while maintaining accuracy.

AI/ML Backend Engineer Intern | *Innovatics, Ahmedabad, Gujarat* — **May 2023 - Jul 2023**

- Implemented a computer vision application using FastAPI and Flask, processing 150+ images/min with 98% accuracy for real-time use cases.
- Applied scalable backend APIs handling 50+ concurrent requests/min, improving system reliability and integration efficiency.
- Optimized backend workflows, reducing response time by 40% and increasing overall processing throughput.

Projects

WildFace Recognition | *Model Inversion Attacks, Black Box Adversarial optimization* — **Sep 2025 - Dec 2025**

- Performed black-box model inversion attacks using NES and autoencoders on 1,288 LFW images, achieving 98.1% confidence and 100% attack success in ∼5K queries.
- Created a modular attack-defense pipeline to evaluate reconstruction quality, query efficiency, and confidence trade offs.
- Programmed defense mechanisms (top-k filtering, noise injection, anomaly detection) achieving 100% detection (1% FPR) and increasing attack cost by 6–7x.

Polymer Structural Energy Prediction System | *Rutgers University (funded project)* — **Feb 2025 - Present**

- Built fine-Tuned ML system to predict polymer energy states from 2,916 CIF structures (156 atomic coordinates for C-H-O bond), enabling identification of low-energy configurations for layer formation.
- Improved features using DFT-based calculations and nearest-neighbor interactions, and developed stacked ensemble models (RF, XGBoost, LightGBM, MLP) improving performance from 73% to 80% MCC (+7%).
- Designed an end-to-end ML pipeline with automated experimentation, hyperparameter tuning, and result visualization through a frontend interface.

Behavioral Security | *Feature Engineering, Fraud Detection, Model Optimization* — **Feb 2025 - Jun 2025**

- Constructed a hybrid fraud detection model combining behavioral biometrics (keystroke, mouse dynamics) with phishing features (∼30 attributes), improving unsafe session detection by 45%.
- Customized all new composite features (e.g., behavioral entropy, deception likelihood, phishing exposure) from 10K+ synthetic sessions, enabling context-aware classification of user behavior.
- Enhanced detection using XGBoost with threshold tuning and class imbalance handling (SMOTE), achieving ∼68–70% recall for fraud cases with improved sensitivity to unsafe interactions.

Publications

Unlocking the Power of Machine Learning in Education | CRC Press — **Mar 2025**
From Threads to Algorithms: AI Innovations in Textile Production | Springer Nature — **Oct 2024**
Review on Sentiment Analysis of Social Media for Stock Market Prediction | Springer India — **Jun 2024**
Empirical Evaluation of Deep Learning Models for Time Series Datasets | Elsevier — **Jan 2023**